

Breaking the Financial Barrier: A Low-Cost Approach to Teaching Virtual Production at Universities

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Figure 1: Students working on their Virtual Production projects.

Abstract

A detailed explanation of the virtual production teaching method used successfully for four years at Technische Hochschule Ostwestfalen-Lippe. We developed an effective approach to teaching virtual production concepts without requiring costly LED volumes and high-end camera tracking hardware. Instead, we primarily use consumer-grade products, such as HTC VIVE trackers, and substitute LED walls with flat-screen TVs.

CCS Concepts

• **Applied computing** → **Media arts; Performing arts; Education**; • **Social and professional topics** → **Adult education**; • **Computing methodologies** → **Computer graphics**.

Keywords

Virtual Production, Realtime Rendering, Education, Cost Efficiency, Technology Integration

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1 Introduction

Virtual production is a filmmaking technique that integrates digital environments and elements with live-action footage in real time. This approach enables filmmakers to visualize and interact with computer-generated imagery (CGI) or virtual sets during the actual filming process, rather than relying on post-production. The technique leverages advanced technologies such as real-time rendering, motion capture, and virtual reality to create immersive and realistic scenes. These scenes are often displayed on large LED walls, allowing the seamless merging of virtual and physical sets during principal photography.

To address the increasing demand for skilled labor in the rapidly advancing field of Virtual Production, numerous universities are establishing dedicated programs. The acquisition, setup, and maintenance of LED walls and professional motion tracking hardware involve substantial financial investments. Furthermore, due to the rapid pace of technological advancements, these multimillion-dollar investments may become obsolete within a span of 3-5 years. Our

approach emphasizes the instruction of foundational concepts in Virtual Production, utilizing primarily cost-effective, consumer-grade hardware to mitigate financial risk while ensuring adaptability to technological changes.

2 Objectives

The primary objectives of this presentation are to:

- (1) Discuss the pedagogical benefits of using cost efficient techniques over high-end hardware
- (2) Showcase practical applications and case studies of our approach.
- (3) Provide educators with strategies for effectively teaching Virtual Production concepts

3 Method

While a comprehensive explanation of the virtual production workflow is beyond the scope of this talk, the main components include:

- a camera with a tracking system to capture its position and focus within the studio
- servers running Unity or Unreal Engine to manage and render the 3D model of the virtual environment in real-time based on the camera tracking data
- a display device for the rendered background. This background can be used as a green-screen live composite or displayed on a projection screen or LED wall during filming, acting as the backdrop to the physical set. This technique is commonly referred to as 'in-camera Visual Effects'.

3.1 Technical Set-Up:

Instead of utilizing a large-scale LED volume, we have established a simplified virtual production setup in our green screen studio. For camera tracking, we employ an HTC VIVE system with three base stations, covering an area of approximately 10 x 5 meters. Our media server consists of a consumer-grade PC equipped with a Nvidia RTX3090 gaming graphics card and a Blackmagic Decklink video I/O card, running Unreal Engine.

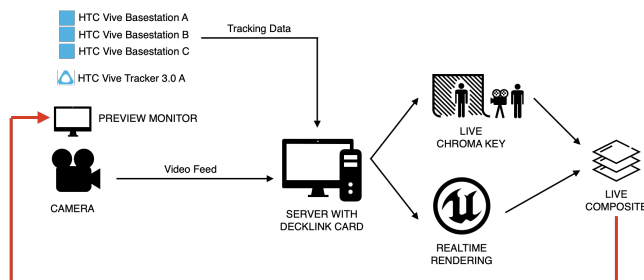


Figure 2: Greenscreen workflow

We also utilize a 60-inch consumer flat-screen TV as an economical alternative to professional LED panels. In addition to using this as a static display, we have mounted an additional HTC VIVE tracker on the TV and developed a method to dynamically change the displayed content based on the camera and screen position. This

approach allows us to compensate for the smaller screen size by adjusting the screen's position relative to the camera frustum.[Meier and Schuh 2022]

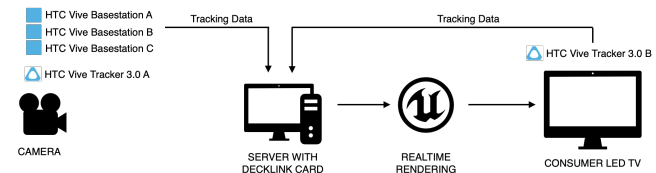


Figure 3: In-Camera VFX workflow

3.2 Teaching method:

Students work in groups of three to five persons, selecting a scene from a movie of their choice to recreate in Unreal Engine. The project is divided into three stages:

3.2.1 Stage 1: Theoretical Foundations. The students are introduced to the fundamental concepts of virtual production in a theoretical manner. This includes an overview of motion capture techniques, real-time render engines, and the structure of the individual components of a LED volume, such as the LED wall, LED controllers, camera and tracking system. Additionally, students learn how to use Unreal Engine to create their selected virtual environment.

3.2.2 Stage 2: Practical Workshop. All students participate in a hands-on workshop in our green screen studio. They learn to:

- Set up camera tracking using the HTC VIVE system.
- Calibrate the camera and lens.
- Perform live green screen keying and real-time compositing of live-action footage with the Unreal Engine render, displaying it on a reference monitor for preview purposes. Subsequently, they are instructed on setting up nDisplay and performing in-camera VFX using the flat-screen TV.

3.2.3 Stage 3: Filming the Project. Students book the green screen studio and set up the virtual production equipment as learned in Stage 2. They then use their virtual environment to recreate the chosen movie scene, selecting the virtual production method based on the shot requirements. Close-up and medium-wide shots are executed with in-camera VFX using the flat-screen TV, while wide shots are accomplished with live green screen keying.

4 Results

Our findings indicate that our method is effective for teaching virtual production concepts in university environments. The hands-on multi-stage approach significantly enhances student engagement, motivation, and information retention. Although our technical setup is not fully suited for creating production ready professional output due to latency issues and the only basic chroma keying algorithm built into Unreal Engine, empirical data reveal positive feedback from both educators and students, citing increased interest and improved understanding of complex concepts. Additionally, students were able to seamlessly apply their knowledge in high-end virtual production environments.

5 Conclusion

Our teaching method has proven successful. By replacing high-end virtual production equipment with low-budget consumer-grade alternatives, we enable universities to effectively teach virtual production concepts without the financial burden of acquiring multimillion-dollar funding. Additionally, the setup is easily scalable and can be established in a regular classroom, eliminating the need for large-scale studios and allowing for the demonstration of fundamental virtual production concepts.

6 Significance

This talk underscores the importance of adapting to financial challenges in education to keep pace with evolving technologies. By

sharing successful implementations and practical tips, we aim to empower educators to enhance their teaching methodologies and enrich the learning experiences of their students.

Acknowledgments

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